

An Evaluation of Deep Learning Models for Sugarcane Leaf Disease Detection

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Abstract—Sugarcane is a significant commodity in many economies, but its yield is constantly threatened by leaf diseases such as red rot, smut, rust, and leaf scald. To identify such issues, farmers have historically depended on skilled visual inspections, a process that is labor-intensive and prone to errors. Recent advances in deep learning (DL) offer feasible alternatives, notably in the area of automated image identification. In order to evaluate the classification of sugarcane leaf images obtained from a Kaggle dataset, this investigation evaluates five prominent convolutional neural network (CNN) architectures: EfficientNetB0, ResNet50, InceptionV3, Xception, and MobileNetV2. Prior to model training, photos were preprocessed by resizing, normalizing pixel intensities, and employing various data augmentation approaches. Afterwards, we trained the networks using industry-standard hyperparameters, such as a 0.001 learning rate for the Adam optimizer and the categorical cross-entropy loss function. Among the models tested, ResNet50 demonstrated superior performance across the five assessed models, reaching the most prominent training accuracy of 93.75% and validation accuracy of 94.00%, as well as the lowest training loss (0.2553) and validation loss (0.2252), indicating exceptional generalization ability. EfficientNetB0 achieved a constant performance level, ranking second with a training accuracy of 90.50% and a validation accuracy of 89% with relatively low loss values. Xception demonstrated consistent generalization with results that were balanced, with a training accuracy of 86.50% and a validation accuracy of 87%. In contrast, MobileNetV2 recorded the training accuracy of 85.75% and the slightly lower validation accuracy of 82.99%, while InceptionV3 reported the lowest training accuracy of 81.25% and the lowest validation accuracy of 77.99%, respectively. The study emphasizes the effectiveness of DL as a potent tool for early disease identification in agriculture, which could lead to practical applications that help farmers react quickly to new dangers.

Index Terms—Sugarcane Leaf Diseases, Deep Learning, CNNs, Transfer Learning.

I. INTRODUCTION

Agricultural biodiversity is crucial for supplying food and raw materials to humanity and is a fundamental element of civilization on Earth [1]. A plant can become infected when pathogenic organisms, including fungi, bacteria, and nematodes, as well as soil pH, temperature fluctuations, air moisture and humidity, and other conditions, persist. Plant diseases can damage plant growth, function, and structure, affecting those who depend on them. Failure to take proper precautions can harm plants and lower product quality, quantity, and production. Thus, plant diseases must be identified and classified early in agriculture [2].

Sugarcane is a globally significant crop that is indispensable to numerous industries, including the production of biofuels and sugar. Many different types of bacteria, fungi, viruses, protozoa, and phytoplasma can infect sugarcane plants. Some of the most notable maladies that affect sugarcane are mosaic disease, ring spot, grassy shoot, and red rot. This red rot, often known as sugarcane cancer, results from the infectious agent *Colletotrichum falcatum* Went. A prevalent sign of mosaic disease in the Co740 population in Maharashtra, India, is decreased fertility. The virus that causes sugarcane mosaics is to blame [3]. Ring Spot primarily affects the older leaves of sugarcane. The causative agent is the fungus *Leptosphaeria sacchari*. Grassy Shoot results from phytoplasma infection. The chlorophyll production in sugarcane leaves is inhibited by grassy shoot disease. White leaf disease (WLD) represents a significant economic challenge within the sugarcane industry, particularly affecting ratoon sugarcane and consequently impacting yield [4]. Phytoplasma, an obligatory plant parasite that harms the phloem tissue, causes WLD. The leafhopper insect vectors transmit. Affected sugarcane leaves have cream-white stripes parallel to the midrib that cover the entire leaf. WLD also causes stunted stalks, without lateral shoots on the top parts, causing plant death. Growers visually monitor disease indications across the field and burn sick crops to prevent spread. This strategy takes time to monitor large sugarcane plantations for diseased areas. Modern computational machine learning (ML) approaches, when applied to precision agriculture technology, have the potential to provide a viable alternative to human-based approaches for detecting sugarcane WLD in the field.

Recently, DL for computer vision (CV) has advanced in object categorization and detection. A number of CV problems have been measured against the Large-Scale Visual Recognition Challenge (ILSVRC) [5], which uses the ImageNet dataset. As of 2012, no state-of-the-art classification model has ever achieved a top-5 error rate of 16.4% on the ImageNet dataset until CNN-based AlexNet DL surpassed all of them.

The goal of this study is to analyze multiple DL models in order to identify a solution that is both accurate and practicable for field deployment. Due to their image processing strengths, we chose EfficientNetB0, ResNet50, InceptionV3, Xception, and MobileNetV2. To find the optimum model for real-world images with different illumination, angles, and quality, we

fine-tuned these designs using a publicly available Kaggle dataset. We use transfer learning to adapt models pre-trained on massive datasets like ImageNet to recognize sugarcane leaf diseases faster and with less data. Our findings demonstrate DL's potential in precision agriculture and offer practical advice for integrating such systems into farmer decision-support tools. Our goal is to provide a scalable, resilient solution for early identification and action, protecting crop yields and promoting sustainable farming.

The rest of the document is structured as follows: Section 2 analyzes the current body of literature on Sugarcane. Section 3 outlines the methodology of these techniques, whereas Section 4 describes the performance evaluation. Section 5 draws a conclusion to the work and delineates potential avenues for forthcoming studies.

II. LITERATURE REVIEW

Sugarcane is a vital agricultural item cultivated in tropical and subtropical regions, renowned for its contributions to sugar production, biofuel generation, and its use in many food and beverage sectors. Its economic significance encompasses various industries, rendering it an essential crop for international commerce and industrial operations. Multiple DL models for finding sugarcane diseases are tested on a collection of over 6,670 photos split into five groups. At 98%, Vision Transformer (ViT) had the highest accuracy, while EfficientNet got to 96.7%. Even though VGG-19 and MobileNet-V2 were also tested, ViT scored the highest [6]. A separate study uses the open-source Sugarcane Leaf Dataset, which contains 6,748 images representing 11 disease categories, to investigate DL approaches for disease identification in sugarcane leaves. The research assesses EfficientNet models (ranging from EfficientNet-b0 to EfficientNet-b7, including EfficientNetv2 variants) and juxtaposes them with alternative models such as ResNetv2-50 and InceptionV4. EfficientNet-b6 attained the best accuracy of 93.39%, closely followed by InceptionV4 at 93.10% [7]. Another investigation assesses the categorization of sugarcane leaf diseases employing the AlexNet DL model, utilizing a dataset including 2,569 images spanning five disease classifications: Healthy, Mosaic, Redrot, Rust, and Yellow. This evaluates the efficacy of two activation functions: ReLU and LeakyReLU. The findings indicate that the AlexNet model utilizing LeakyReLU attained the best accuracy of 90.67%, whilst ReLU reached 87.90% [8]. A different research investigation investigates DL methodologies for the recognition and categorization of sugarcane leaf diseases, utilizing a dataset of 13,842 photos of both healthy and sick leaves. CNN was employed to classify the photos, with an accuracy of 95% for healthy leaves. Findings show that DL models help in early disease diagnosis, which in turn helps farmers cut down on crop losses [9]. Another research investigation uses a dataset consisting of 2,940 images divided into six categories (five diseased and one healthy) to investigate DL models for sugarcane disease detection. The evaluated models comprise VGG-19, ResNet-34, and ResNet-50, with ResNet-50 attaining the maximum accuracy of 93.20% when

trained on all layers and assessed with augmentation. The research investigates object detection with Faster-RCNN and YOLOv3, attaining mean average precisions (mAP) of 58.30% and 54.11%, respectively [10]. In a further investigation, an Attention-based Multi-level Residual CNN (AMRCNN) classifies sugarcane leaf disorder using 2,569 images from five categories. AMRCNN accuracy reached 86.53%, exceeding VGG19, ResNet50, and EfficientNet-B7 models. To assist farmers in detecting infections early, an Android app allowed real-time diagnosis [11]. An alternative article delves into a technique for sugarcane disease detection using ML and image processing, including topics like Eyespot, Leaf Scald, Yellow Leaf, and Pokkah Boeng. This system is able to classify diseases because it makes use of Case-Based Reasoning (CBR), Neuro-Fuzzy systems, and Artificial Neural Networks (ANN). With a 95% accuracy rate for Eyespot sickness, Neuro-Fuzzy and CBR models were the best of the bunch [12]. One further study used CNN for disease detection with a dataset of 3,256 sugarcane photos split into nine categories (eight diseased and one healthy). Its accuracy of 95.61% was higher than that of KNN, RNN, and SVM models [13]. Using the Environmental Adaptation Method (EEAM) on a dataset with four disease classes, another study trained a CNN tailored for sugarcane infection detection. With an accuracy of 89.12%, it outperforms current models like VGG16 and VGG19, and it brings attention to the potential for future integration with mobile apps for real-time analysis [14]. The additional research uses the Sugarcane Leaf Dataset, which has 6,748 images covering eleven illness classifications, with DL models like EfficientNet-B7 and DenseNet201. EfficientNet-B7 showed that DL may enhance the identification of sugarcane diseases with an accuracy of 99.79% [15].

III. RESEARCH METHODOLOGY

This section gives a synopsis of the approach used to develop and assess DL models for sugarcane leaf disease detection. This process includes the following fundamental phases: data collection, data description, data preparation, and model evaluation.

A. Data Description

The dataset encompasses a total of 600 images, with 500 designated for training purposes and 100 allocated for testing. The training set is structured into five distinct subfolders, with each subfolder corresponding to a specific class: Healthy, Rust, Red Rot, Mosaic, and Yellow. Each subfolder contains around 100 images, providing a uniform distribution across the various classes. The effectiveness of the model is assessed with a test set that includes 20 photos for every class.

B. Data Preprocessing

The dataset has been prepared for training by scaling the images to 224×224 pixels, which is the size that the chosen CNN architectures need for input. For efficient training, the pixel intensities were tuned to lie within the interval [0, 1]. The training set was enhanced with data augmentation

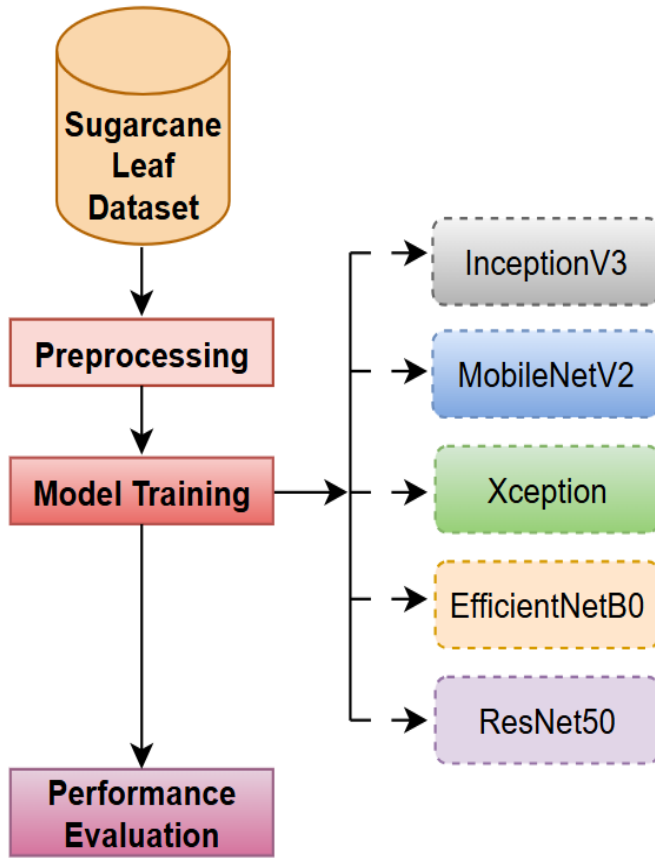


Fig. 1: Workflow diagram.

procedures such as random rotations (up to 20 degrees), horizontal flips (up to 20 percent), zooms (up to 20%), and brightness modifications (0.8-1.2) in order to reduce overfitting and increase model generalizability.

C. Model Development

Figure 2 depicts the workflow for this study. The initial phase entails collecting data from Kaggle, followed by pre-processing techniques including scaling, normalization, and augmentation. Five CNN models—EfficientNetB0, ResNet50, InceptionV3, Xception, and MobileNetV2—were seeded with ImageNet weights and further refined using transfer learning methodologies. The training procedure implemented the Adam optimizer with a learning rate of 0.001, a batch size of 32, and a loss function of categorical cross-entropy. It ran for 20 epochs. We used early stopping to end training after five epochs if the validation loss didn't improve. The evaluation of performance encompassed metrics including loss and accuracy. The training procedure employed TensorFlow on a GPU-enabled system. The models were evaluated using the test set, with metrics carefully documented for both training and validation phases to evaluate performance.

D. EfficientNetB0

EfficientNetB0 is a CNN that serves as the foundation of the EfficientNet series [16]. The core of the design is a compound

scaling technique that evenly scales the network's depth, width, and resolution with respect to a set of specified scaling factors. Mobile Inverted Bottleneck Convolutions (MBConv) are the foundational principles of EfficientNetB0. Compared to conventional convolutions, MBConv blocks use less processing since they are composed of depthwise separable convolutions. Adaptively recalibrating feature maps to increase representational strength, the network also uses squeeze-and-excitation (SE) blocks and swish activation functions. Initially, the architecture includes a stem block for processing, which is then followed by multiple MBConv blocks. Finally, for output, there is a global average pooling layer and a softmax classifier.

E. InceptionV3

The InceptionV3 model is an advanced CNN that improves upon its predecessors, the InceptionV1 and V2, by implementing various significant enhancements that boost performance and efficiency in computation. The "Inception module," used by the design, is a series of convolutional filters of varying sizes (1x1, 3x3, 5x5) that are applied in parallel, and then their outputs are combined. Because of its architecture, the network can detect features at different sizes without incurring a corresponding spike in computing expense. Factorized convolutions are new to InceptionV3 and allow for the reduction of parameters by breaking down bigger filters (like 5x5) into smaller ones (like two 3x3 convolutions). Aside from using global average pooling to boost generalization and decrease overfitting, the model also has fully linked layers. Training also makes use of auxiliary classifiers to stabilise and improve gradient flow [17].

F. MobileNetV2

MobileNetV2 is a lightweight CNN architecture for mobile and embedded vision applications that improves efficiency and accuracy over MobileNetV1. The "Inverted Residuals" block, which expands the input using a depthwise separable convolution with a narrow bottleneck layer, a lightweight depthwise convolution, and projected output to the desired output dimension, is MobileNetV2's main innovation. By decreasing parameters while keeping rich feature representations, this structure speeds computation. To reduce computing complexity, MobileNetV2 uses linear bottleneck layers with linear activations in each block's final layer. This model works well for mobile and resource-constrained applications since it balances accuracy and computing cost with lightweight depthwise separable convolutions and linear bottlenecks [18].

G. ResNet50

The ResNet50 architecture is a deep CNN that is a member of the ResNet (Residual Networks) series. It was developed to resolve the vanishing gradient issue that is commonly encountered in large deep networks. It has 50 layers and presents the idea of residual learning, which allows the network to learn residual mappings rather than direct mappings by adding shortcut connections to bypass one or more layers. Training very deep models is made easier by these shortcut

connections, which make gradients move more readily across the network during backpropagation. A sequence of residual blocks, each with two or three convolutional layers, batch normalization, and ReLU activations comprise the ResNet50 architecture. The efficiency of the model is further maximized by the incorporation of bottleneck layers, which lower the amount of parameters and calculations. Because ResNet50 can efficiently train deep networks, it has demonstrated notable performance gains in picture classification applications, especially on datasets like ImageNet [19].

H. Xception

The Xception architecture utilizes depthwise separable convolutions, which is a significant change compared to the Inception framework. Xception is a deep CNN that expands upon the Inception architecture. In depthwise separable convolutions, the filter is employed to each input channel individually in the first stage, and then the outputs are combined in the second stage. This is in contrast to traditional convolutions, which apply a filter to all input channels. The number of parameters and computational cost are both reduced significantly through this separation, without sacrificing performance. A more effective and scalable model is formed by Xception, which substitutes depthwise separable convolutions for the conventional Inception modules. The structure is made up of fourteen "modules" that contain thirty-six convolutional layers, with each module having a regular convolution layer after a sequence of depthwise separable convolutions [20].

IV. RESULTS AND DISCUSSION

Table I delineates a comparative analysis of the effectiveness of five distinct DL models, namely EfficientNetB0, ResNet50, InceptionV3, Xception, and MobileNetV2, with regard to key parameters such as training accuracy, validation accuracy, training loss, and validation loss.

TABLE I: Comparison of Applied Models

Model	Train Acc.	Val. Acc.	Train Loss	Val. Loss
InceptionV3	81.25%	77.99%	0.5393	0.5666
MobileNetV2	85.75%	82.99%	0.5102	0.6190
Xception	86.50%	87.00%	0.4130	0.4566
EfficientNetB0	90.50%	89.00%	0.2782	0.3874
ResNet50	93.75%	94.00%	0.2553	0.2252

With a training accuracy of 93.75% and a validation accuracy of 94.00%, ResNet50 comes out as the best performer in both training and validation. It achieved the maximum level of accuracy. The fact that it has the lowest training loss (0.2553) and validation loss (0.2252) demonstrates that the technique is not only successful in learning the data but also works effectively with data that it has not previously considered. In light of this, it appears that ResNet50 is the most trustworthy model among those that were evaluated, since it operates exceptionally well in both training and inference. Following closely behind is EfficientNetB0, which also has a validation accuracy of 89% and a training accuracy of 90.50%. Because it maintains a relatively low training loss(0.2782) and

a fair validation loss(0.3874), it has a balanced performance, which implies that it has strong performance with minimal overfitting. Even while it does not quite reach the level of ResNet50 or EfficientNetB0, Xception performs pretty well, with a training accuracy of 86.50% and validation accuracy of 87%. This demonstrates that it has strong generalization in its performance. With a training accuracy of 85.75% and a validation accuracy of 82.99%, MobileNetV2 is also a solid model; however, it is slightly behind in terms of their accuracy. Although it has a greater training loss (0.5102) and validation loss (0.6190) compared to the other techniques, this indicates that it may have a slightly more difficult time learning the data and generalizing its findings. InceptionV3 has the lowest training accuracy of 81.25% and the lowest validation accuracy of 77.99%, making it the least accurate of the three models in this comparison. Moreover, it has the largest training loss (0.5393) and validation loss (0.5666), which indicates that the model has difficulty learning properly. Both of these losses are relatively high. It is abundantly evident that ResNet50 is the most dependable model, as it provides the optimal balance between accuracy and loss. On the other hand, InceptionV3 would need additional tuning and changes in order to optimize its performance on this particular task.

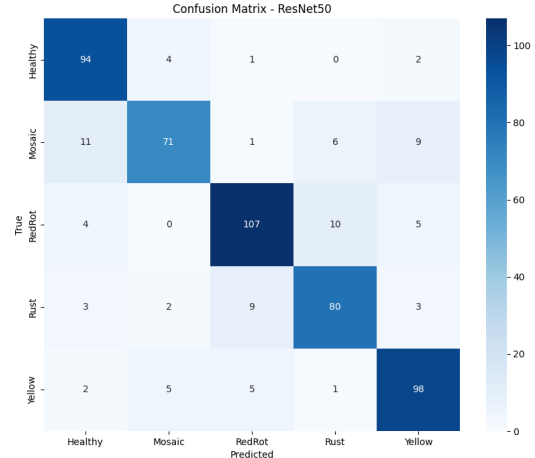


Fig. 2: Confusion-matrix of ResNet50.

In Figure 2, the performance of the ResNet50 model is satisfactory, particularly in terms of categorizing the Healthy (94 correct) and Yellow (98 correct) categories. RedRot is correctly classified (107 times), but at the same time, it is incorrectly labeled as Yellow ten times. However, mosaic is sometimes mistaken with Healthy (11) and Yellow (9), despite the fact that it has 71 proper classifications. There are a few incorrect categories into other classes, but Rust displays a total of eighty accurate classifications. The model's overall efficiency is quite good; nevertheless, it has difficulty discriminating between diseases that are quite similar to one another, such as Mosaic, Red Rage, and Yellow.

V. CONCLUSION AND FUTURE WORK

This work presents a comparative analysis of DL models for the detection of sugarcane leaf diseases. The research assesses five notable CNNs—EfficientNetB0, ResNet50, InceptionV3, Xception, and MobileNetV2—utilizing a dataset including 600 sugarcane images. To enhance model generality, data preprocessing techniques included image scaling, normalization, and augmentation. Experimental results indicated that ResNet50 attained the highest performance, achieving a validation accuracy of 94%, followed by EfficientNetB0 at 89% and Xception at 87%. MobileNetV2 and InceptionV3 had relatively inferior performance. ResNet50 had the most favorable training and validation loss metrics, signifying its proficiency in generalizing to novel data.

Further models will be built and assessed to enhance accuracy beyond present benchmarks, focusing on designs that better capture sugarcane leaf disease complexity. Additionally, these models could be linked to mobile smartphone apps to help farmers diagnose leaf diseases by capturing photos, especially in remote places without expert resources. In addition, IoT devices may monitor crops in real time for disease identification and control. More diverse datasets, ensemble approaches for improved performance, and explainability techniques to increase transparency and user confidence will be focus of future study.

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